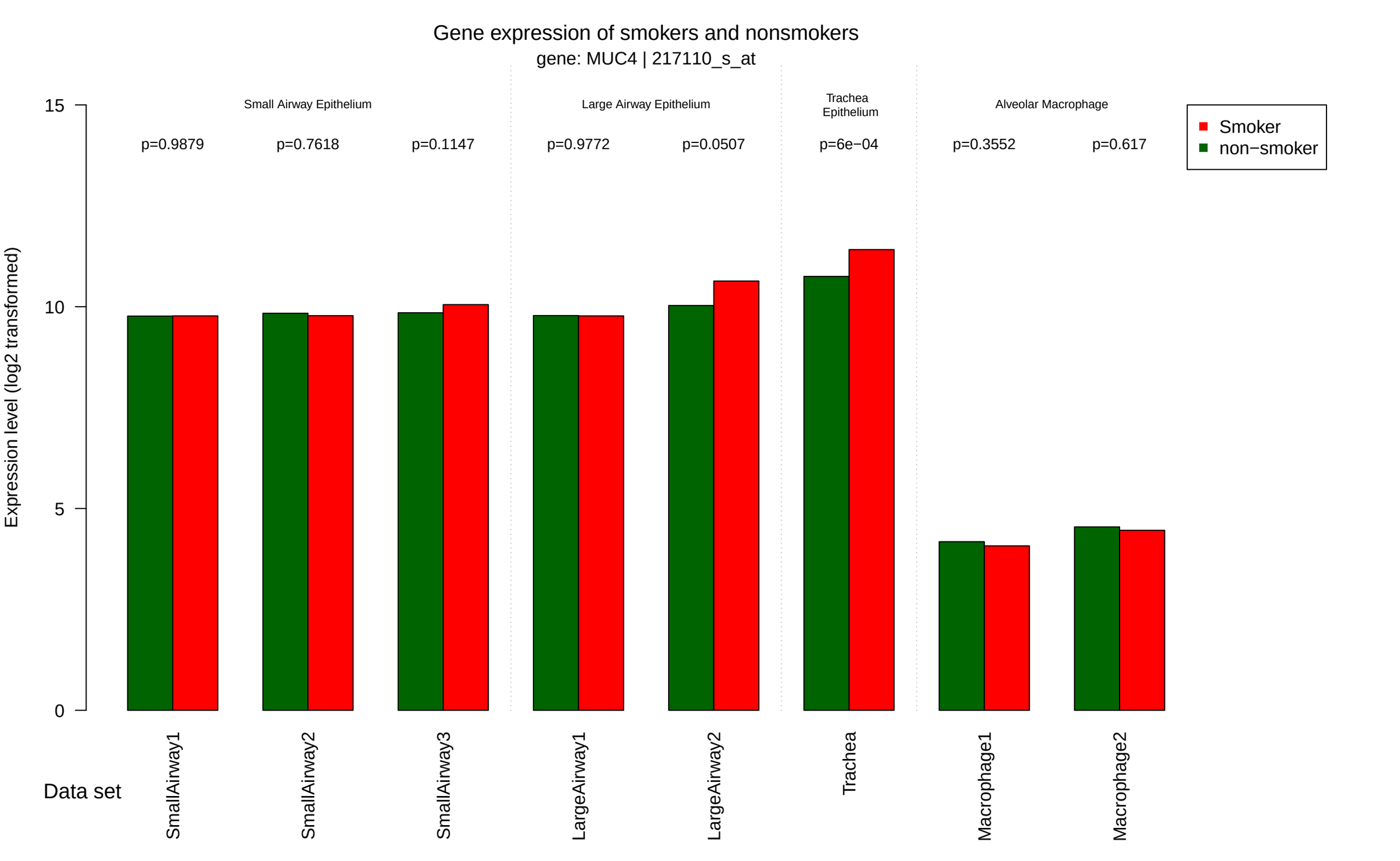
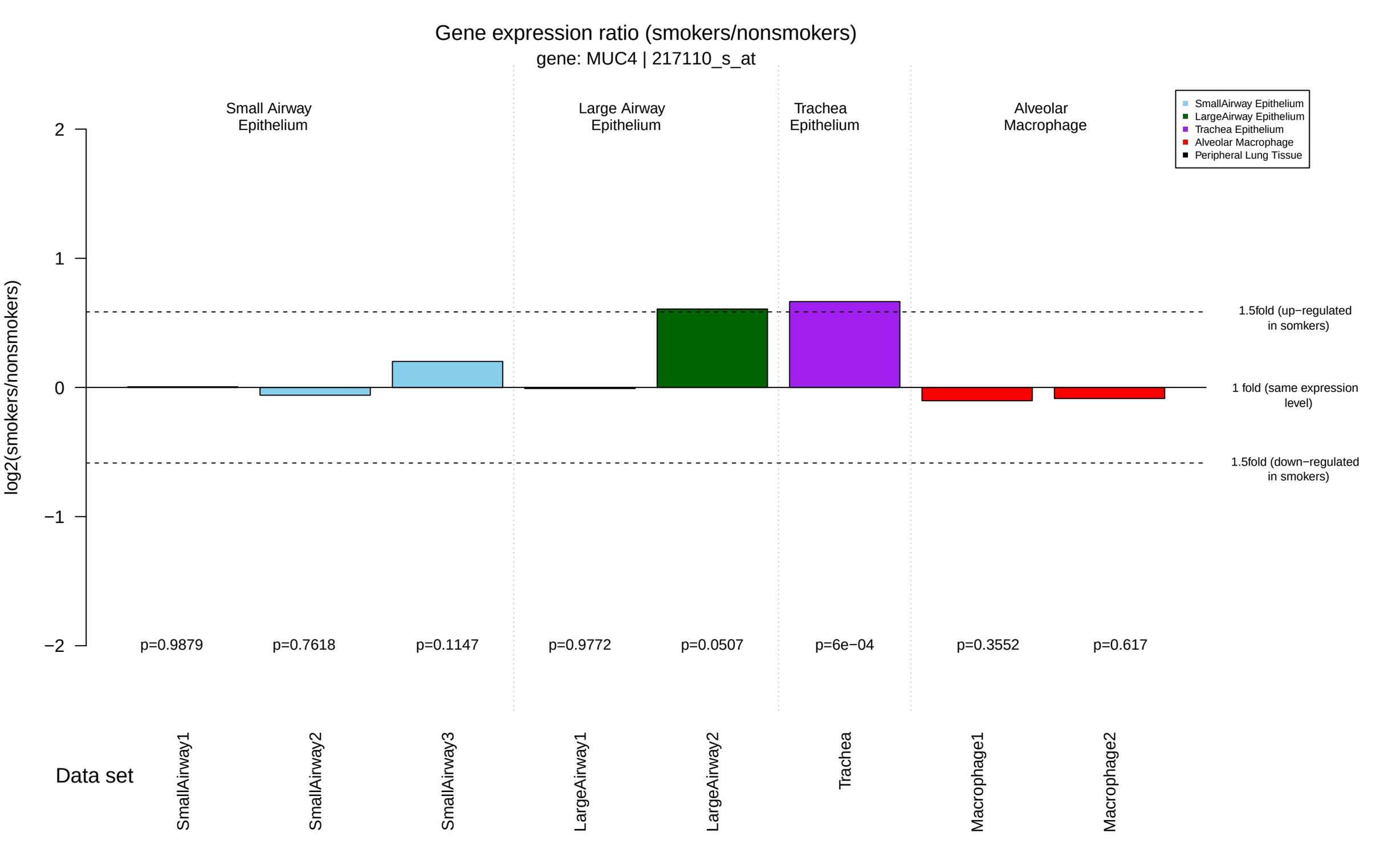
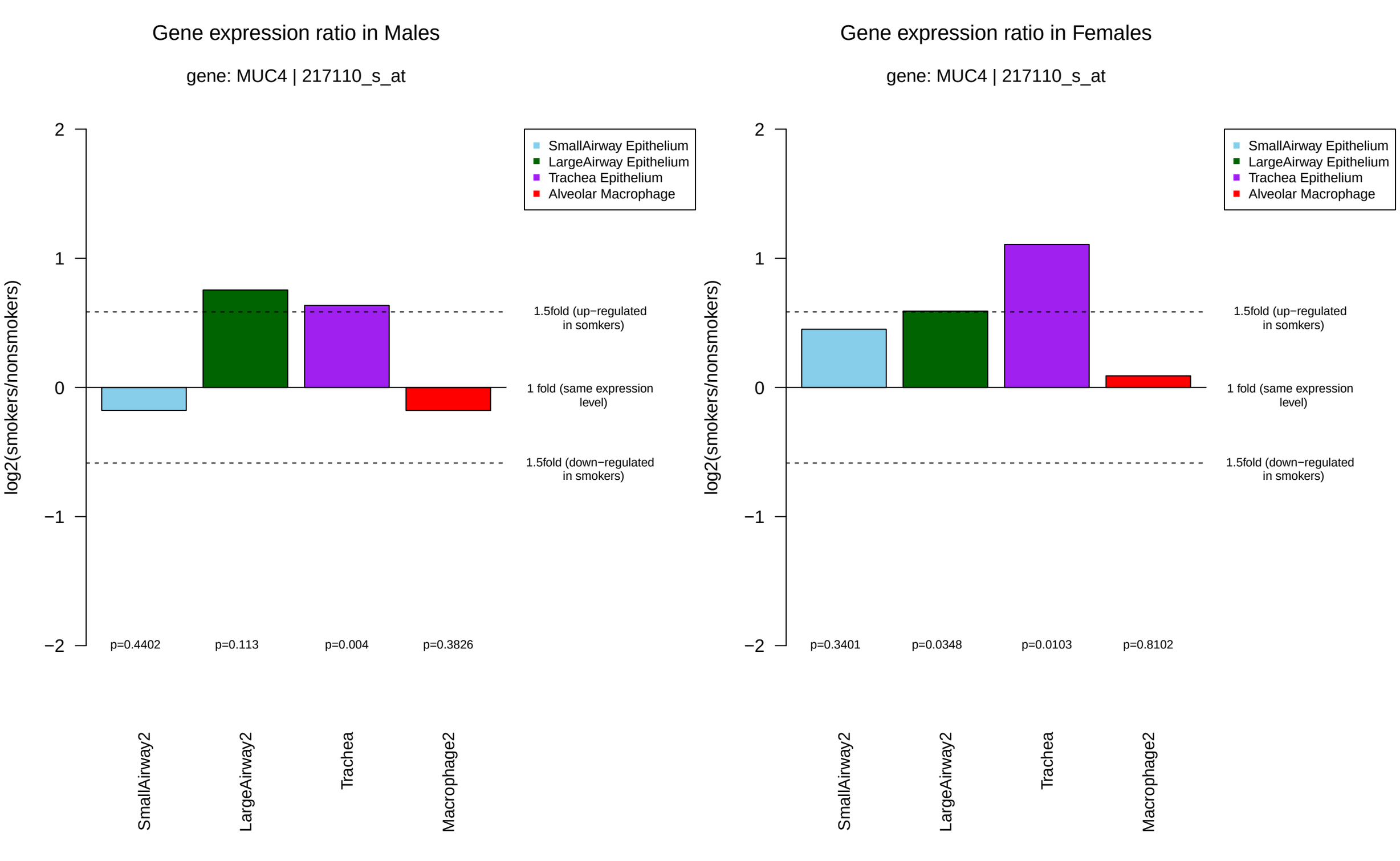
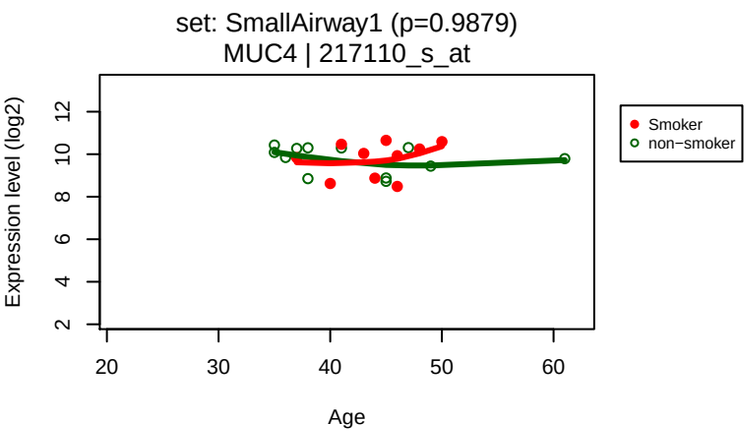
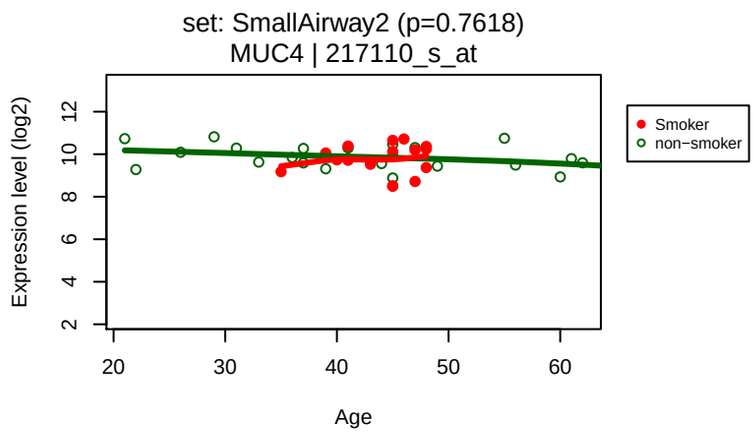
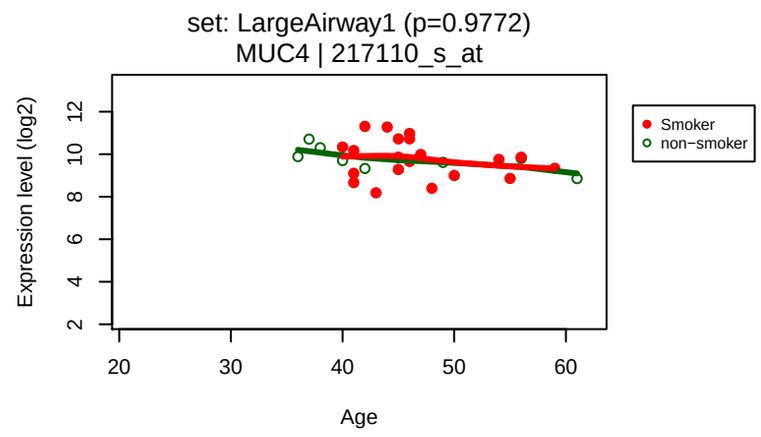

Gene expression of smokers and nonsmokers
gene: MUC4 | 217110_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9879
p=0.7618
p=0.1147
p=0.9772
p=0.0507
p=6e−04
p=0.3552
p=0.617
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

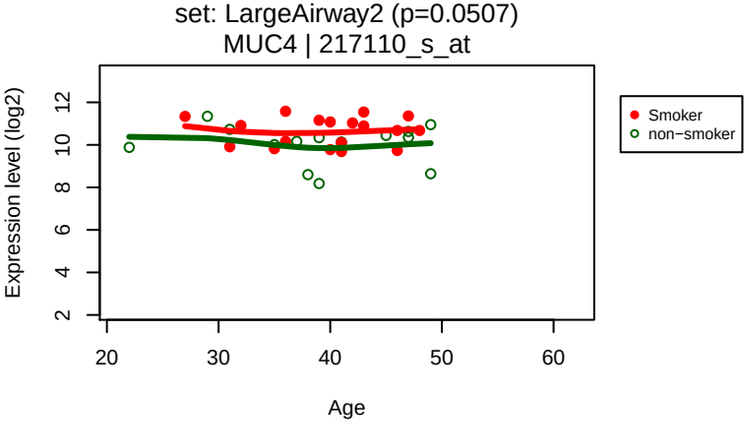

Gene expression ratio (smokers/nonsmokers)
gene: MUC4 | 217110_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9879
p=0.7618
p=0.1147
p=0.9772
p=0.0507
p=6e−04
p=0.3552
p=0.617
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

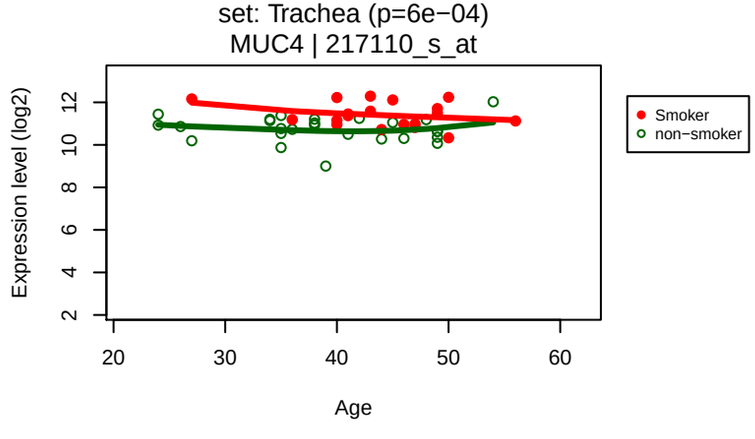

Gene expression ratio in Males
gene: MUC4 | 217110_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4402
p=0.113
p=0.004
p=0.3826
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MUC4 | 217110_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3401
p=0.0348
p=0.0103
p=0.8102
SmallAirway2
LargeAirway2
Trachea
Macrophage2

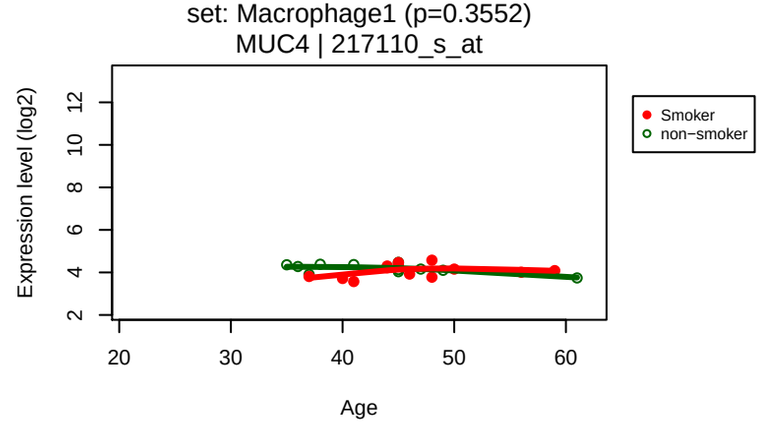

set: SmallAirway1 (p=0.9879)
MUC4 | 217110_s_at
Expression level (log2)
Age
Smoker
non−smoker

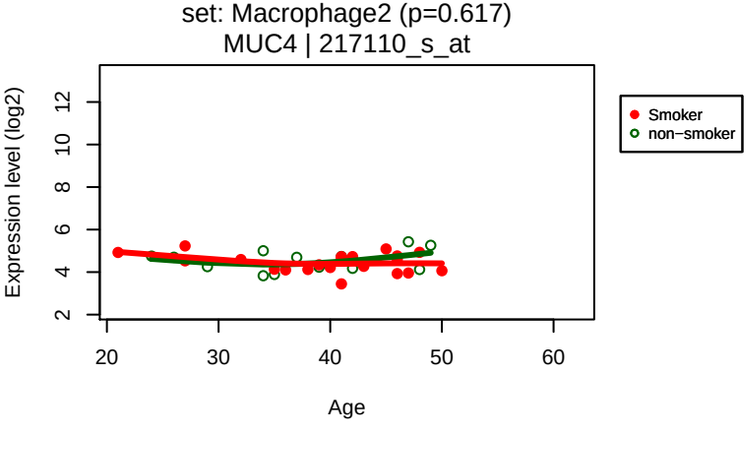

set: SmallAirway2 (p=0.7618)
MUC4 | 217110_s_at
Expression level (log2)
Age
Smoker
non−smoker

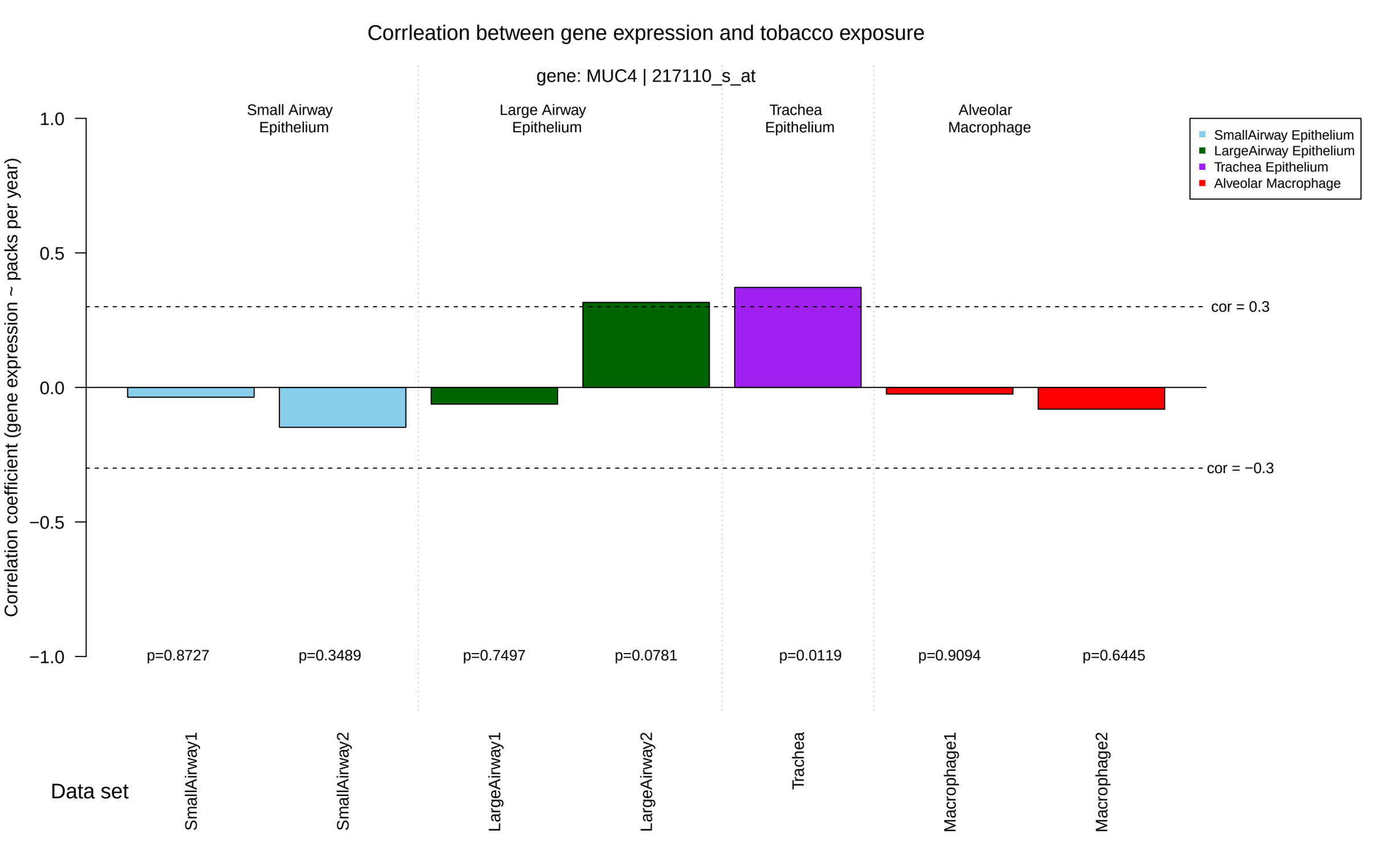

set: LargeAirway1 (p=0.9772)
MUC4 | 217110_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0507)
MUC4 | 217110_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=6e−04)
MUC4 | 217110_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3552)
MUC4 | 217110_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.617)
MUC4 | 217110_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MUC4 | 217110_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8727
p=0.3489
p=0.7497
p=0.0781
p=0.0119
p=0.9094
p=0.6445
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2